

Will start at 9:32 am

ECON1050 Tutorial 3

Functions of One Variable

Semester 2 2026

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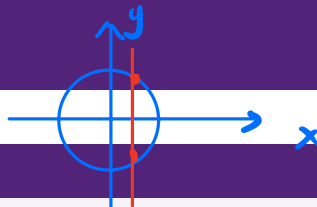
Announcement

- ▶ Quiz 1 has been released. It covers topics 1 and 2.
- ▶ Due date: 21/08/2026 4:00 pm.
- ▶ It is worth 8% of your final grade!

Today's Plan

- ▶ Recap: key ideas from Lecture 3
- ▶ Tutorial questions: Q1–Q5

Recap: Functions



Definition

A function of a real variable x with domain D is a rule that assigns a unique real number to each $x \in D$. The notation is

$$y = f(x).$$

Here x is the independent variable or argument, and y is the dependent variable.

Domain and range

- ▶ The **domain** is the set of all admissible values of x .
- ▶ The **range** is the set of all resulting values $f(x)$.

Domain $\xrightarrow{x} \boxed{f(\cdot)} \xrightarrow{f(x)} \text{Codomain}$

$$\text{Range} = \{f(x) : x \in D\}$$

Vertical line test

The graph of a function has at most one point of intersection with any vertical line.

key point: f assigns one input exactly one output

Recap: Coordinate, Domain, and Monotonicity

Coordinate system

A point on a graph is represented by an ordered pair (x, y) .

Common restrictions

- ▶ A denominator cannot be zero.
- ▶ The expression inside a square root must be non-negative. $\sqrt{\quad}$
- ▶ A logarithm is only defined for a positive argument. $\ln(\cdot)$

Increasing and Decreasing Functions

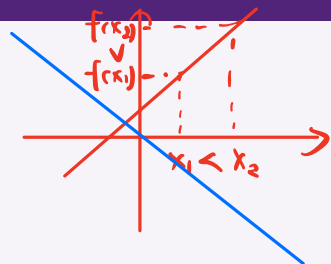
For $x_1, x_2 \in I$ with $x_1 < x_2$:

Increasing: $f(x_1) \leq f(x_2)$,

Strictly increasing: $f(x_1) < f(x_2)$,

Decreasing: $f(x_1) \geq f(x_2)$,

Strictly decreasing: $f(x_1) > f(x_2)$.



Recap: Linear Functions

General form

$$y = ax + b.$$

The coefficient a is the slope and b is the intercept on the y -axis.

Slope from two points

Given two points (x_1, y_1) and (x_2, y_2) with $x_1 \neq x_2$,

$$a = \frac{y_2 - y_1}{x_2 - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$$

Point-slope formula

If the slope is a and the line passes through (x_1, y_1) , then

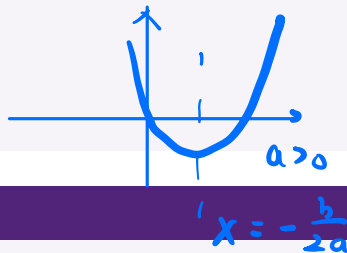
$$y - y_1 = a(x - x_1).$$

Recap: Quadratic Functions

General form

$$y = ax^2 + bx + c.$$

A quadratic function has a parabola-shaped graph.



Roots and vertex

- ▶ Roots are found by solving $ax^2 + bx + c = 0$.
- ▶ The x-coordinate of the vertex is

$$x = -\frac{b}{2a}.$$

Recap: Other Function Families

Polynomial functions

A polynomial of degree n has the form

$$y = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0.$$

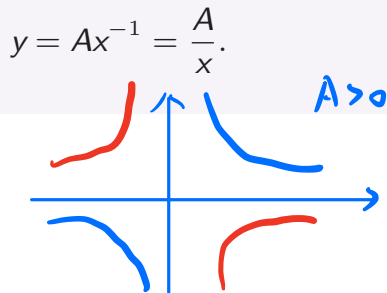
Power and hyperbolic functions

Power functions have the form $y = Ax^a$. A simple hyperbolic function has the form

$$y = A \cdot x^a$$

↓
-1

$$y = \frac{A}{x}$$



Recap: Exponential and Logarithmic Functions

Exponential functions

An exponential function has a variable in the exponent:

$$y = ba^x \quad \text{or} \quad f(t) = Aa^t.$$



Natural exponential and logarithms

$$y = e^x, \quad \ln(ab) = \ln a + \ln b, \quad \ln\left(\frac{a}{b}\right) = \ln a - \ln b, \quad \ln(x^c) = c \ln x.$$

$$y = \ln x, \quad x > 0$$

Recap: New Functions and Inverse Functions

New functions from old functions

Given two functions $f(x)$ and $g(x)$, we can form

$$f(x) + g(x), \quad f(x) - g(x), \quad f(x)g(x), \quad \frac{f(x)}{g(x)}, \quad f(g(x)).$$

composition

$$h(x) = f(g(x))$$

$$= \sin(x^2)$$

$$x^2: g(x)$$

$$\sin x: f(x)$$

Inverse functions

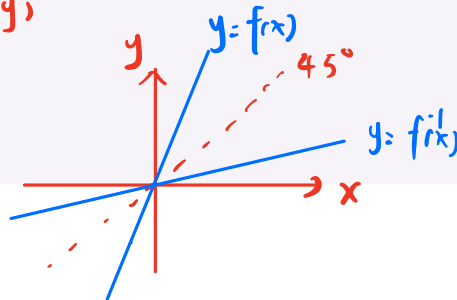
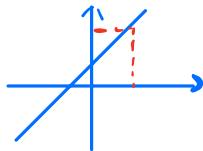
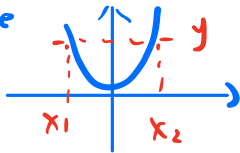
If f is one-to-one, its inverse g satisfies

$$y = f(x) \iff x = g(y)$$

$$g(y) = x \iff y = f(x).$$

The graphs of inverse functions are symmetric about the line $y = x$.

Example



Recap: Economic Functions

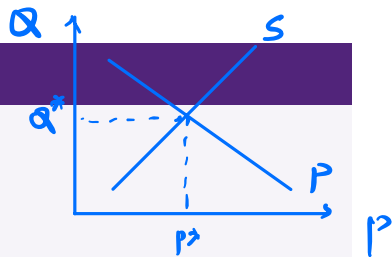
Demand and supply

Demand and supply can be written as functions of price:

$$Q_D = D(P), \quad Q_S = S(P).$$

How many Quant
Consumer want to buy
given P

How many Quant
Producers want to sell, given P



Inverse functions in economics

In economic diagrams, price is often placed on the vertical axis. Therefore we often use inverse functions:

$$P_D = d(Q), \quad P_S = s(Q).$$

Revenue

If price is a function of quantity, then total revenue is

$$TR(Q) = P(Q)Q.$$

Question 1

Let

$$F(x) = 1 + \frac{4x}{x^2 + 4}.$$

4 mins

- (a) Compute $F(0)$, $F(-2)$, $F(2)$ and $F(3)$.
- (b) What happens to $F(x)$ when x becomes large and positive or large and negative?
- (c) Draw a rough sketch of the graph of F .

$$(a) \quad F(0) = 1 + \frac{4 \cdot 0}{0^2 + 4} = 1$$

$$F(-2) = 1 + \frac{4 \cdot (-2)}{(-2)^2 + 4} = 1 - 1 = 0$$

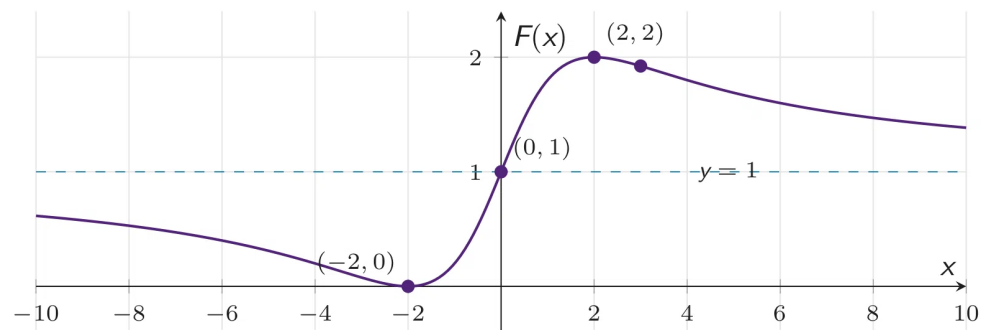
$$F(2) = 1 + \frac{4 \cdot 2}{2^2 + 4} = 1 + 1 = 2$$

$$F(3) = 1 + \frac{4 \cdot 3}{3^2 + 4} = \frac{25}{13}$$

$$c_b) F(x) \rightarrow ? \cdot \begin{cases} x \rightarrow \infty \\ x \rightarrow -\infty \end{cases}$$

$$\textcircled{1} x \rightarrow \infty, F(x) = 1 + \frac{4x}{x^2+4} \xrightarrow{x \rightarrow \infty} 1 + \frac{4}{\frac{4}{x} + \frac{4}{x}} \rightarrow 1 + \frac{4}{\infty+0} \rightarrow 1+0=1$$

$$\textcircled{2} x \rightarrow -\infty, F(x) = 1 + \frac{4x}{x^2+4} = 1 + \frac{4}{\frac{x}{1} + \frac{4}{x}} \xrightarrow{x \rightarrow -\infty} 1 + \frac{4}{-\infty+0} = 1 - \frac{4}{\infty} = 1$$



Tutorial Q1: Knowledge used

Knowledge used

- ▶ A function assigns a unique value $F(x)$ to each admissible value of x .
- ▶ Function values are found by substituting the given x values into the rule.
- ▶ The long-run behaviour of a rational expression is studied by comparing numerator and denominator as x becomes large.
- ▶ A graph can be sketched using key points and asymptotic behaviour.

Tutorial Q1: Function values

Solution

$$F(0) = 1 + \frac{4(0)}{0^2 + 4} = 1.$$

$$F(-2) = 1 + \frac{4(-2)}{(-2)^2 + 4} = 1 - \frac{8}{8} = 0.$$

$$F(2) = 1 + \frac{4(2)}{2^2 + 4} = 1 + \frac{8}{8} = 2.$$

$$F(3) = 1 + \frac{4(3)}{3^2 + 4} = 1 + \frac{12}{13} = \frac{25}{13}.$$

Tutorial Q1: Large x behaviour

Solution

Rewrite the second term as

$$\frac{4x}{x^2 + 4} = \frac{4}{x + \frac{4}{x}}.$$

When x is large and positive,

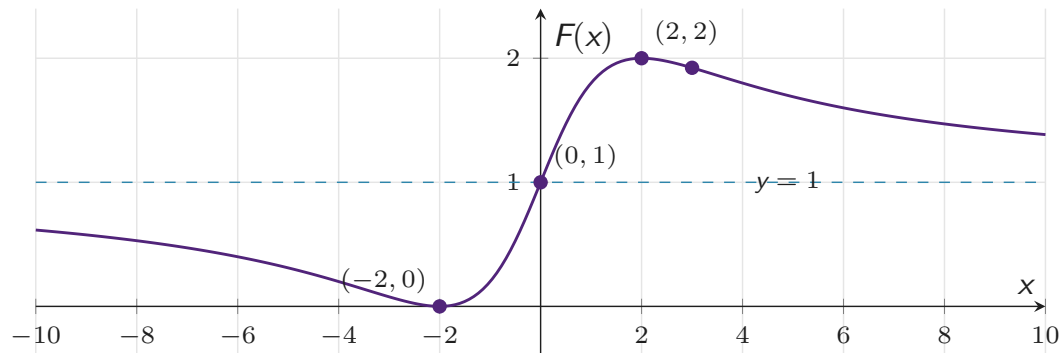
$$\frac{4}{x + \frac{4}{x}} \approx 0^+, \quad F(x) \rightarrow 1 \text{ from above.}$$

When x is large and negative,

$$\frac{4}{x + \frac{4}{x}} \approx 0^-, \quad F(x) \rightarrow 1 \text{ from below.}$$

Therefore, $y = 1$ is the horizontal asymptote.

Tutorial Q1: Rough sketch

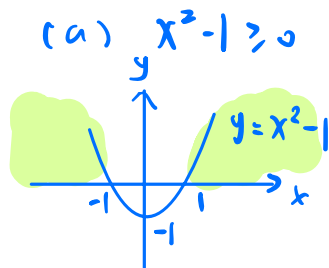


4 mins

Question 2

Find the domain of the following functions. Use a sign diagram to help define the domains where appropriate.

(a) $f(x) = \sqrt{x^2 - 1}$, (b) $g(x) = \frac{2x - 1}{x^2 - x}$.



$$x \in (-\infty, -1] \cup [1, \infty)$$

(b) $x^2 - x \neq 0$
 $x(x - 1) \neq 0$
 $x \neq 0, 1$
 $x \in \mathbb{R} \setminus \{0, 1\}$

Tutorial Q2: Knowledge used

Knowledge used

- ▶ The domain is the set of all admissible x values.
- ▶ For a square root, the argument must be non-negative.
- ▶ For a fraction, the denominator cannot be zero.
- ▶ A sign diagram helps solve inequalities such as $x^2 - 1 \geq 0$.

Tutorial Q2(a): Domain of f

Solution

For

$$f(x) = \sqrt{x^2 - 1},$$

we require

$$x^2 - 1 \geq 0.$$

Factor:

$$x^2 - 1 = (x + 1)(x - 1).$$

The critical points are $x = -1$ and $x = 1$.

Interval	$(-\infty, -1)$	$(-1, 1)$	$(1, \infty)$
$(x + 1)(x - 1)$	+	-	+

Hence

$$D_f = (-\infty, -1] \cup [1, \infty).$$

Tutorial Q2(b): Domain of g

Solution

For

$$g(x) = \frac{2x-1}{x^2-x} = \frac{2x-1}{x(x-1)},$$

the denominator must not be zero:

$$x(x-1) \neq 0.$$

Therefore,

$$x \neq 0, \quad x \neq 1.$$

So the domain is

$$D_g = \mathbb{R} \setminus \{0, 1\}$$

or equivalently

$$D_g = (-\infty, 0) \cup (0, 1) \cup (1, \infty).$$

Tutorial Q3

Question 3

4 mins

(a) Find the slope of the straight line

$$3x + 4y = 12.$$

(b) Determine the equation for the straight line L_1 passing through $(-3, 5)$ and $(2, 7)$.

(a) General form: $y = ax + b$. $3x + 4y = 12 \Leftrightarrow 4y = 12 - 3x$
↳ slope $y = \frac{12}{4} - \frac{3}{4}x = 3 - \frac{3}{4}x$ slope

(b) $y = ax + b = \frac{2}{5}x + \frac{31}{5}$
 $\begin{cases} (-3, 5): 5 = -3a + b & \textcircled{1} \\ (2, 7): 7 = 2a + b & \textcircled{2} \end{cases}$

$$\begin{aligned} \textcircled{2} - \textcircled{1}: 7 - 5 &= 2a + 3a + 0 \\ 2 &= 5a \\ a &= \frac{2}{5} \end{aligned}$$

$$\begin{aligned} \text{Plug } a = \frac{2}{5} \text{ into } \textcircled{2}: 7 &= 2 \cdot \frac{2}{5} + b \\ b &= \frac{31}{5} \end{aligned}$$

Tutorial Q3: Knowledge used

Knowledge used

- ▶ A linear function can be written as $y = ax + b$, where a is the slope.
- ▶ The slope from two points is

$$a = \frac{y_2 - y_1}{x_2 - x_1}.$$

- ▶ The point-slope formula is

$$y - y_1 = a(x - x_1).$$

Tutorial Q3(a): Slope

Solution

Start with

$$3x + 4y = 12.$$

Solve for y :

$$4y = 12 - 3x,$$

$$y = 3 - \frac{3}{4}x.$$

In the form $y = ax + b$,

$$a = -\frac{3}{4}, \quad b = 3.$$

Therefore the slope is

$$-\frac{3}{4}.$$

Tutorial Q3(b): Equation of L_1

Solution

The line passes through $(-3, 5)$ and $(2, 7)$. The slope is

$$a = \frac{7 - 5}{2 - (-3)} = \frac{2}{5}.$$

Using the point-slope formula with $(x_1, y_1) = (2, 7)$,

$$y - 7 = \frac{2}{5}(x - 2).$$

Hence

$$y = \frac{2}{5}x - \frac{4}{5} + 7 = \frac{2}{5}x + \frac{31}{5}.$$

Therefore,

$$L_1 : y = \frac{2}{5}x + \frac{31}{5}.$$

Tutorial Q4

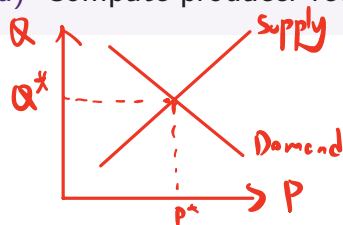
Question 4

Consider the demand and supply curves

5 mins

$$Q_D = 150 - \frac{1}{2}P, \quad Q_S = 20 + 2P.$$

- (a) Find the equilibrium price P^* and quantity Q^* .
- (b) Derive the inverse demand and inverse supply functions.
- (c) Suppose a tax of \$2 per unit is imposed on the producer. How does it influence the equilibrium price?
- (d) Compute producer revenue before and after the tax.



$$\begin{cases} Q = 150 - \frac{1}{2}P & \text{D} \\ Q = 20 + 2P & \text{S} \end{cases}$$

$$0 = 20 - 150 + 2P + \frac{1}{2}P$$
$$P^* = 52$$

$$\text{Plugging } P^* = 52 \text{ into D: } Q^* = 150 - \frac{1}{2} \cdot 52 = 124$$

$$(Q^*, P^*) = (124, 52)$$

$$(b) \quad Q_D = 150 - \frac{1}{2}P \Leftrightarrow \frac{1}{2}P = 150 - Q_D \Leftrightarrow P = 300 - 2Q_D : \text{inverse demand}$$

$$Q_S = 20 + 2P \Leftrightarrow -2P = 20 - Q_S \Leftrightarrow P = -10 + \frac{1}{2}Q_S : \text{inverse supply}$$

(c) For producer, price is the money she can get by selling one unit.

tax of 2 per unit $\Rightarrow$ producer earns \$2 less

$$\Rightarrow P - 2$$

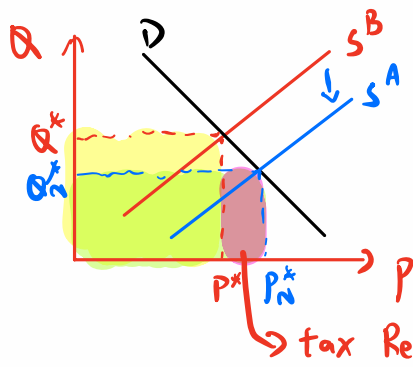
$$Q_S = 20 + 2P \Rightarrow \text{new } Q_S = 20 + 2(P - 2) = 16 + 2P$$

$$\begin{cases} Q_D = 150 - \frac{1}{2}P & \text{①} \\ Q_S = 16 + 2P & \text{②} \end{cases} \quad \text{①} - \text{②}: 0 = 150 - 16 - \frac{1}{2}P - 2P$$

$$P^* = 53.6$$

$$\text{plug } P^* = 53.6 \text{ into ①: } Q^* = 150 - \frac{1}{2} \cdot 53.6 = 123.2$$

$$(Q_N^*, P_N^*) = (123.2, 53.6)$$



$$\text{tax Revenue} = \text{tax} \cdot Q_N^*$$

$$(d) \quad \text{Revenue} = P^* \cdot Q^*$$

$$\text{Before} : P^* \cdot Q^* = 52 \cdot 124 = 6448$$

$$\text{After} : P_N^* \cdot Q_N^* = 53.6 \cdot 123.2 = 6603.52$$

$$\text{tax Revenue} = 2 \cdot 123.2 = 246.4$$

$$\text{Actual Revenue} = P_N^* \cdot Q_N^* - \text{tax Revenue} = 6603.52 - 246.4 = 6357.12$$

Tutorial Q4: Knowledge used

Knowledge used

- ▶ Equilibrium occurs when demand equals supply: $Q_D = Q_S$.
- ▶ Demand and supply can be written as functions of price.
- ▶ Inverse demand and inverse supply write price as a function of quantity.
- ▶ Total revenue is price times quantity:
$$TR = PQ.$$
- ▶ A producer tax shifts the supply curve because the producer receives the consumer price minus tax.

Tutorial Q4(a): Equilibrium

Solution

At equilibrium,

$$Q_D = Q_S = Q^*.$$

So

$$150 - \frac{1}{2}P^* = 20 + 2P^*.$$

Rearrange:

$$130 = 2P^* + \frac{1}{2}P^* = \frac{5}{2}P^*.$$

Thus

$$P^* = \frac{130}{5/2} = 52.$$

Then

$$Q^* = 150 - \frac{1}{2}(52) = 124.$$

Tutorial Q4(b): Inverse functions

Solution

For demand:

$$Q_D = 150 - \frac{1}{2}P.$$

Then

$$\frac{1}{2}P = 150 - Q_D, \quad P = 300 - 2Q_D.$$

For supply:

$$Q_S = 20 + 2P.$$

Then

$$2P = Q_S - 20, \quad P = \frac{1}{2}Q_S - 10.$$

Tutorial Q4(c): Producer tax

Solution

With a \$2 producer tax, if the consumer price is P_c , the producer receives

$$P_c - 2.$$

The new supply curve is

$$Q_S^{\text{tax}} = 20 + 2(P_c - 2) = 16 + 2P_c.$$

New equilibrium:

$$150 - \frac{1}{2}P_c = 16 + 2P_c.$$

Hence

$$134 = \frac{5}{2}P_c, \quad P_c = 53.6.$$

The new quantity is

$$Q^{\text{tax}} = 150 - \frac{1}{2}(53.6) = 123.2.$$

The producer receives $53.6 - 2 = 51.6$ per unit.

Tutorial Q4(d): Producer revenue

Solution

Before tax, producer revenue is

$$R^* = P^* Q^* = 52 \times 124 = 6448.$$

After tax, consumer spending is

$$P_c Q^{\text{tax}} = 53.6 \times 123.2 = 6603.52.$$

Tax revenue is

$$T = 2 \times 123.2 = 246.4.$$

Therefore producer revenue after tax is

$$R^{\text{tax}} = 6603.52 - 246.4 = 6357.12.$$

The decrease in producer revenue is

$$6448 - 6357.12 = 90.88.$$

Tutorial Q5

Question 5

Find the equation of a parabola passing through the three points

$$(1, -3), \quad (0, -6), \quad (3, 15).$$

General form: $y = ax^2 + bx + c = 2x^2 + x - 6$

$$\begin{cases} (1, -3): -3 = a + b + c & \textcircled{1} \\ (0, -6): -6 = 0 + 0 + c & \textcircled{2} \\ (3, 15): 15 = 9a + 3b + c & \textcircled{3} \end{cases} \Rightarrow \begin{cases} 3 = a + b & \textcircled{4} \\ 7 = 3a + b & \textcircled{5} \end{cases}$$

$$c = -6$$

$$\textcircled{5} - \textcircled{4}: 4 = 2a$$

$$a = 2$$

plug $a=2$ into $\textcircled{4}: 3 = 2 + b$
 $b = 1$

Tutorial Q5: Knowledge used

Knowledge used

- ▶ A quadratic function has the form

$$y = ax^2 + bx + c.$$

- ▶ A point (x, y) lies on the graph if substituting its coordinates satisfies the equation.
- ▶ Three points give three equations for the three unknown coefficients a , b and c .

Tutorial Q5: Set up equations

Solution

Let the parabola be

$$y = ax^2 + bx + c.$$

Substitute the three points.

$$(1, -3) : \quad -3 = a + b + c,$$

$$(0, -6) : \quad -6 = c,$$

$$(3, 15) : \quad 15 = 9a + 3b + c.$$

Using $c = -6$,

$$a + b = 3, \quad 9a + 3b = 21.$$

Divide the second equation by 3:

$$3a + b = 7.$$

Tutorial Q5: Solve for the coefficients

Solution

We have

$$a + b = 3, \quad 3a + b = 7.$$

From the first equation,

$$a = 3 - b.$$

Substitute into the second equation:

$$3(3 - b) + b = 7,$$

$$9 - 2b = 7,$$

$$b = 1.$$

Then

$$a = 3 - b = 2, \quad c = -6.$$

Therefore,

$$y = 2x^2 + x - 6.$$

Key takeaways

- ▶ Function evaluation means substituting values into the function rule.
- ▶ Domains are determined by restrictions such as non-negative square-root arguments and non-zero denominators.
- ▶ Linear functions are identified using slopes and points.
- ▶ Market equilibrium is found from $Q_D = Q_S$, and inverse functions express price as a function of quantity.
- ▶ A quadratic passing through three points can be found by solving for a , b and c in $y = ax^2 + bx + c$.